

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE CODE: ITA 06 COURSE NAME: MACHINE LEARNING COURSE

OUTCOME COVERED

CO2: *Neural Network Representation, Perceptron Model, Multilayer Perceptrons, Backpropagation Algorithm, Deep Neural Networks, Genetic Algorithms, Hypothesis Space Search, Genetic Programming, Models of Evaluation and Learning (BL1, BL2)*

Assessment Tool Weightage: 20%

QUESTIONS: 15 Total Marks:100 Annexure A

SCENARIO-BASED QUESTIONS

Code of Conduct:

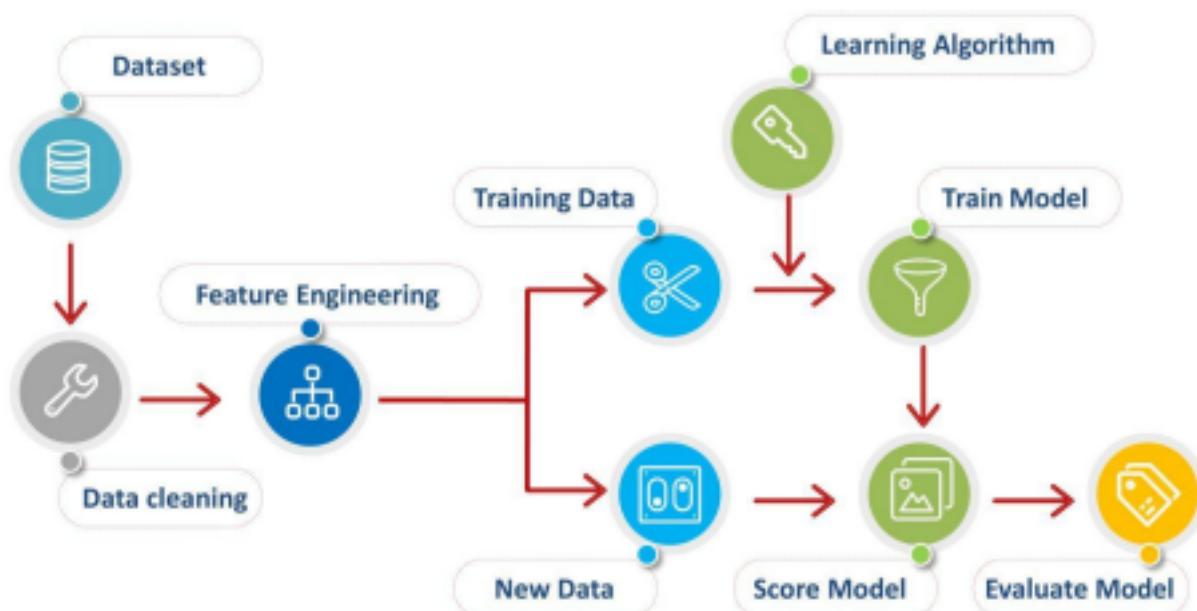
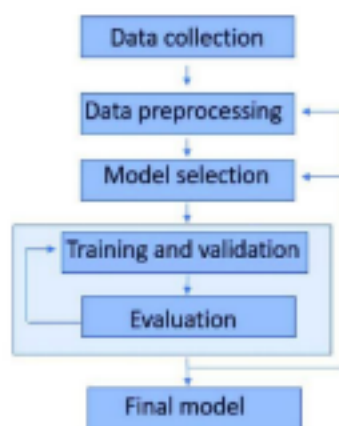
I SANTHOSH/192324146 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

<i>Criteria</i>	<i>Conceptual Understanding</i>	<i>Accuracy of Answers</i>	<i>Application of Knowledge</i>	<i>Completeness of Response</i>	<i>Clarity & Presentation</i>	<i>Total</i>
<i>Weightage</i>	30%	20%	20%	15%	15%	
<i>Q1</i>						
<i>Q2</i>						
<i>Q3</i>						

Signature of the student: Total Marks Awarded SANTHOSH B

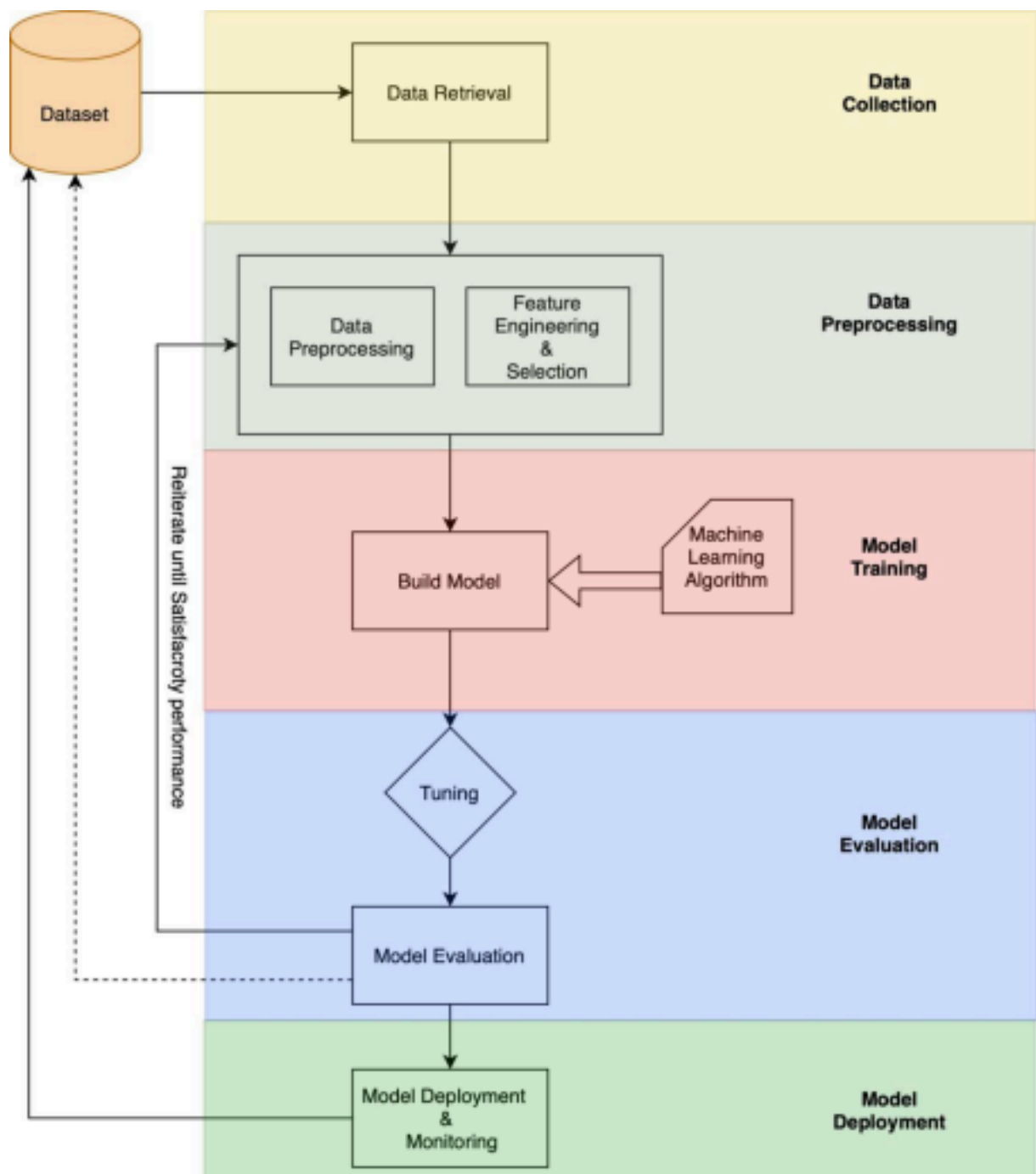
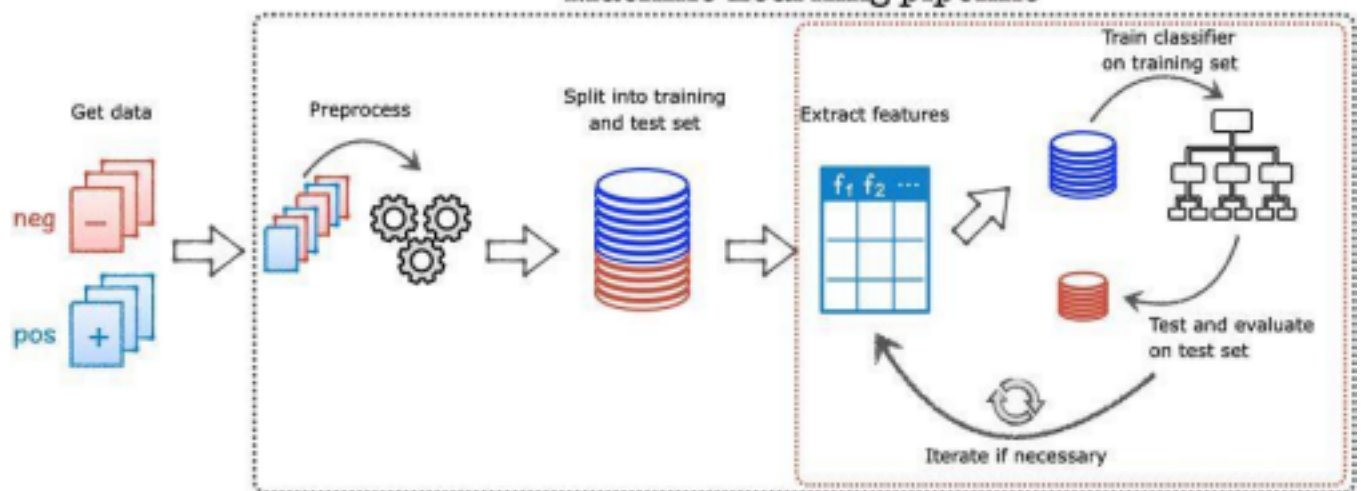
1. With the help of a neat sketch, explain the different functional blocks present in a machine learning system, including data collection, preprocessing, feature extraction, model training, and evaluation. Illustrate the working of the system by providing suitable examples for both supervised and unsupervised learning approaches, clearly highlighting how labelled and unlabelled data are handled within the learning process.



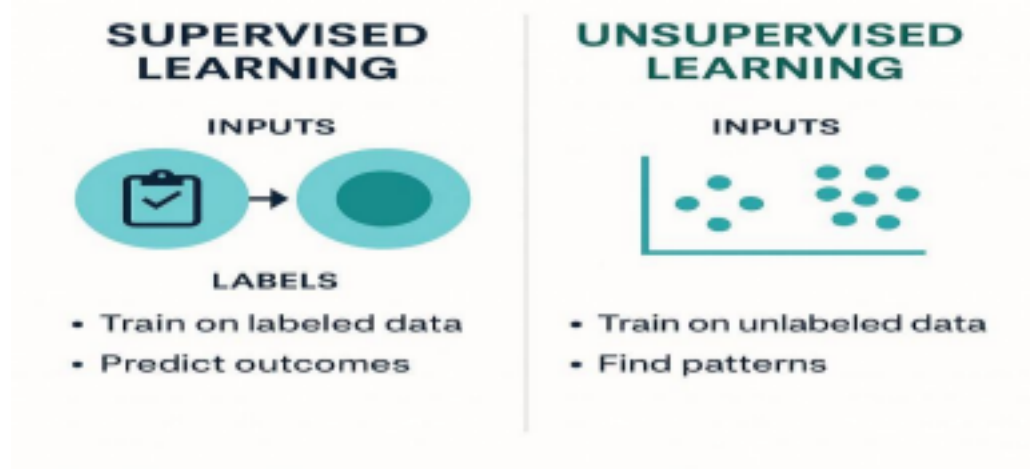
SUPERVISED VS. UNSUPERVISED LEARNING



Machine Learning pipeline



SUPERVISED VS UNSUPERVISED LEARNING



Explanation of Functional Blocks

1. Data Collection

- Collects data from databases, sensors, IoT devices, websites, or user inputs.
- Example: Collecting student attendance and exam scores.

2. Data Preprocessing

- Removes missing values and duplicate records.
- Normalizes data and converts categorical values into numerical form.
- Example: Filling missing attendance values.

3. Feature Extraction/Selection

- Selects important attributes that influence prediction.
- Reduces unnecessary data and improves accuracy.
- Example: Selecting age, blood pressure, and sugar level for disease prediction.

4. Model Training

- The algorithm learns patterns from training data.
- Examples: Decision Tree, Random Forest, Neural Network, SVM.

5. Model Evaluation

- Measures model performance using Accuracy, Precision, Recall, F1-score, etc.
- If performance is poor, retraining is performed.

Supervised Learning

- Uses labelled data.
- Each input has a known output.

Example: Predicting whether a student passes or fails using attendance and study hours.

Input → Study Hours, Attendance

Output Label → Pass / Fail

Unsupervised Learning

- Uses unlabelled data.
- Finds hidden patterns or groups automatically.

Example: Customer segmentation based on purchasing behavior.

Input → Customer Purchase Data

Output → Similar Customer Groups (Clusters)

Difference

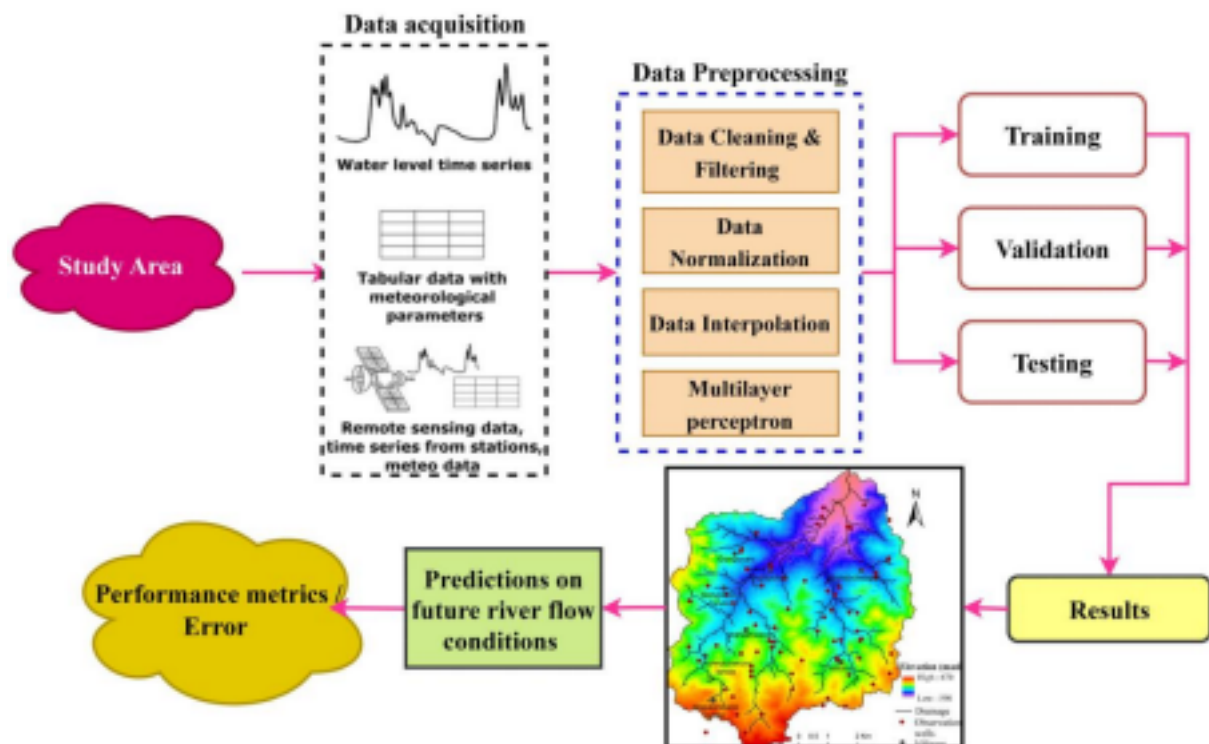
Supervised Learning Unsupervised Learning

Uses labelled data Uses unlabelled data

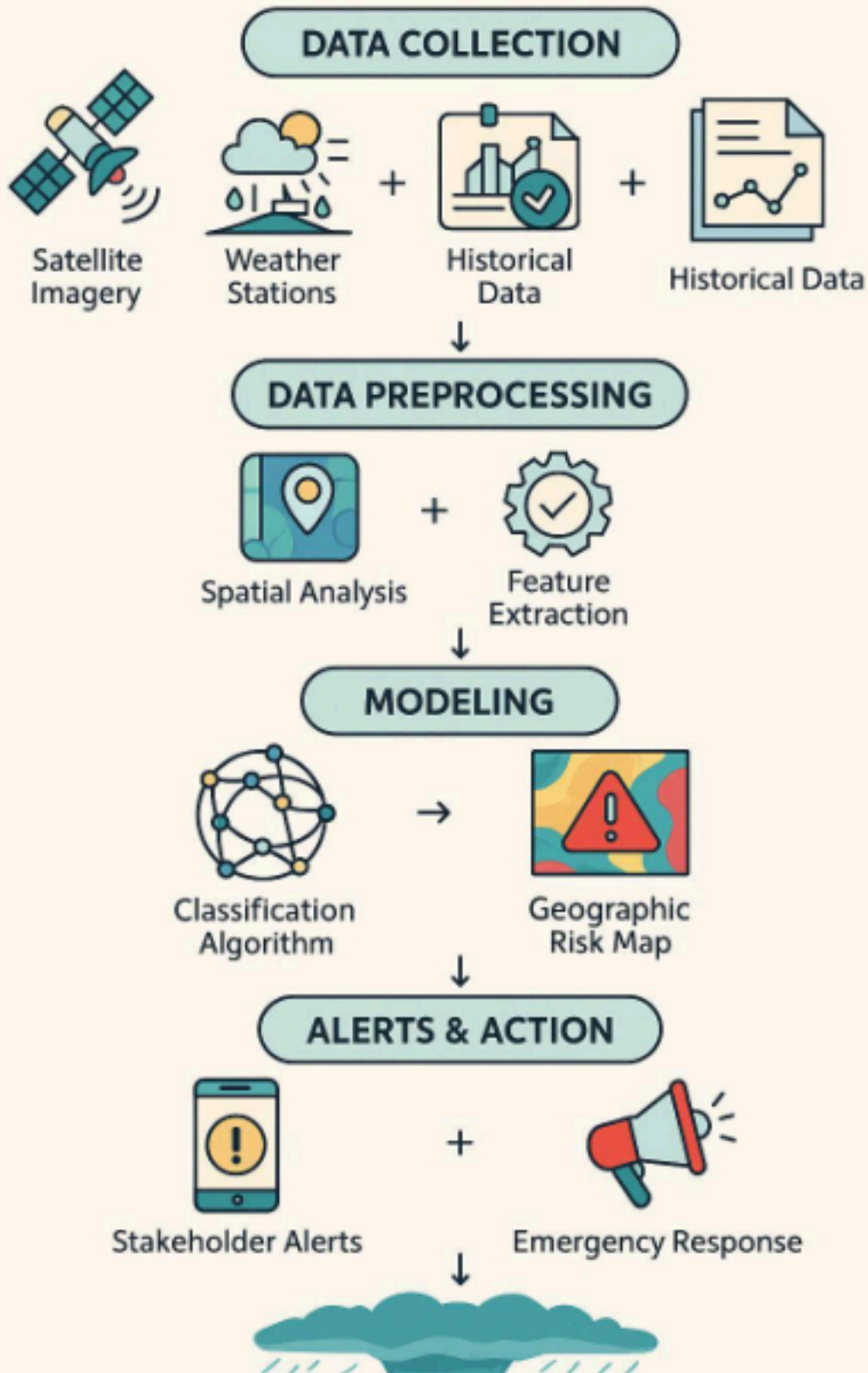
Predicts output Finds hidden patterns

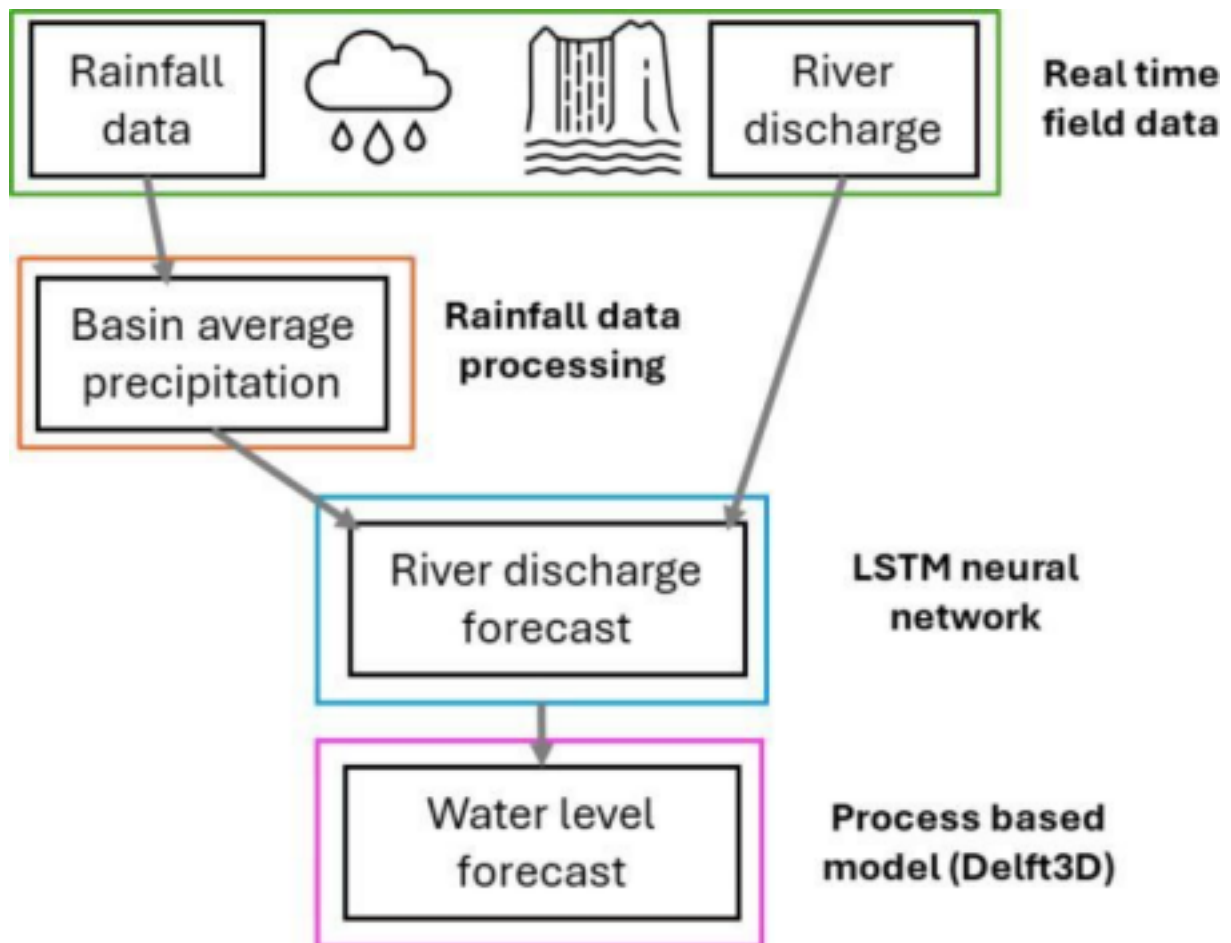
Example: Email Spam Detection Example: Customer Segmentation

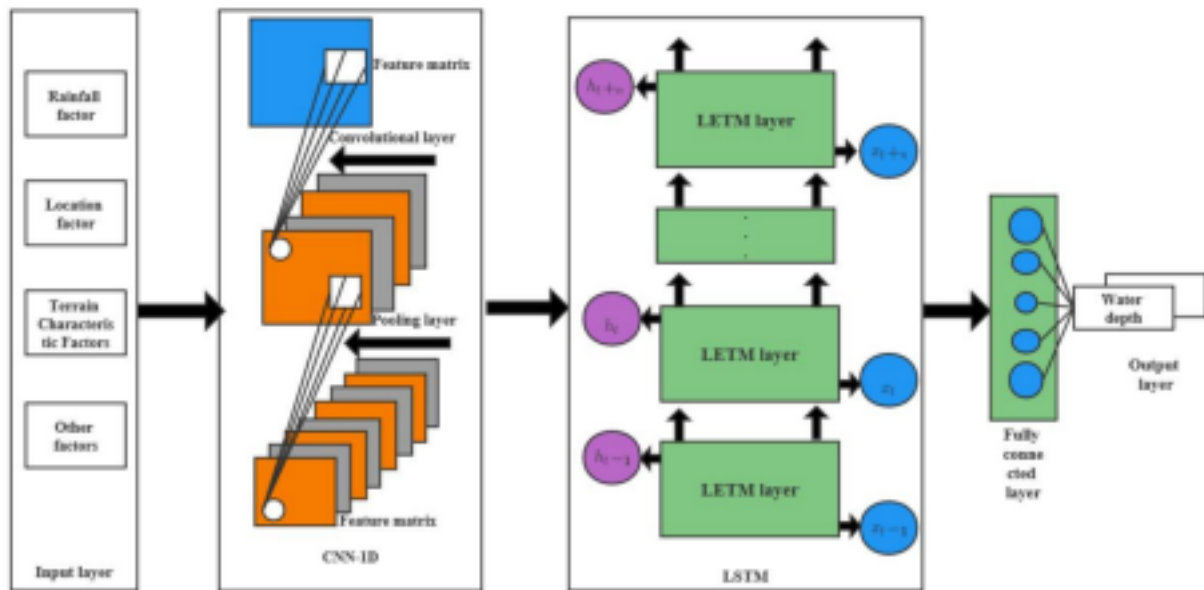
2. Design a machine learning system for disaster management to predict events such as floods and earthquakes using both historical and real-time data sources. Explain the selection of an appropriate model, including whether a neural network or any other algorithm is suitable for this task, along with justification. Describe the data preprocessing steps and feature selection techniques required to improve model performance. Further, explain the complete process of training, validating, and evaluating the model, and discuss how such a system can assist authorities in early warning, risk assessment, and effective emergency response.



PREDICTIVE ANALYSIS FOR NATURAL DISASTER MANAGEMENT







Disaster Management System

Historical Data + Real-Time Sensor Data



Data Preprocessing



Feature Selection



Machine Learning Model



Prediction (Flood/Earthquake)



Early Warning System



Emergency Response

Data Sources

- Historical flood records
- Earthquake records
- Weather data
- River water level sensors
- Rainfall measurements
- Satellite images
- Seismic sensor data

Suitable Model

Artificial Neural Network (ANN)

Reason:

- Learns complex nonlinear relationships.
- Handles large datasets efficiently.
- Suitable for weather and seismic prediction.

Other suitable algorithms:

- Random Forest
- XGBoost
- LSTM (for time-series prediction)

Data Preprocessing

- Remove missing values.
- Remove duplicate records.
- Normalize numerical values.

- Handle noisy sensor readings.
- Encode categorical variables.

Feature Selection

Important features include:

- Rainfall
- River water level
- Temperature
- Humidity
- Soil moisture
- Seismic wave magnitude
- Earthquake depth

Feature selection methods:

- Correlation Analysis
- Principal Component Analysis (PCA)
- Information Gain

Training Process

1. Collect historical data.
2. Split into training (80%) and testing (20%).
3. Train the neural network.
4. Validate using validation data.
5. Tune hyperparameters.
6. Evaluate model performance.

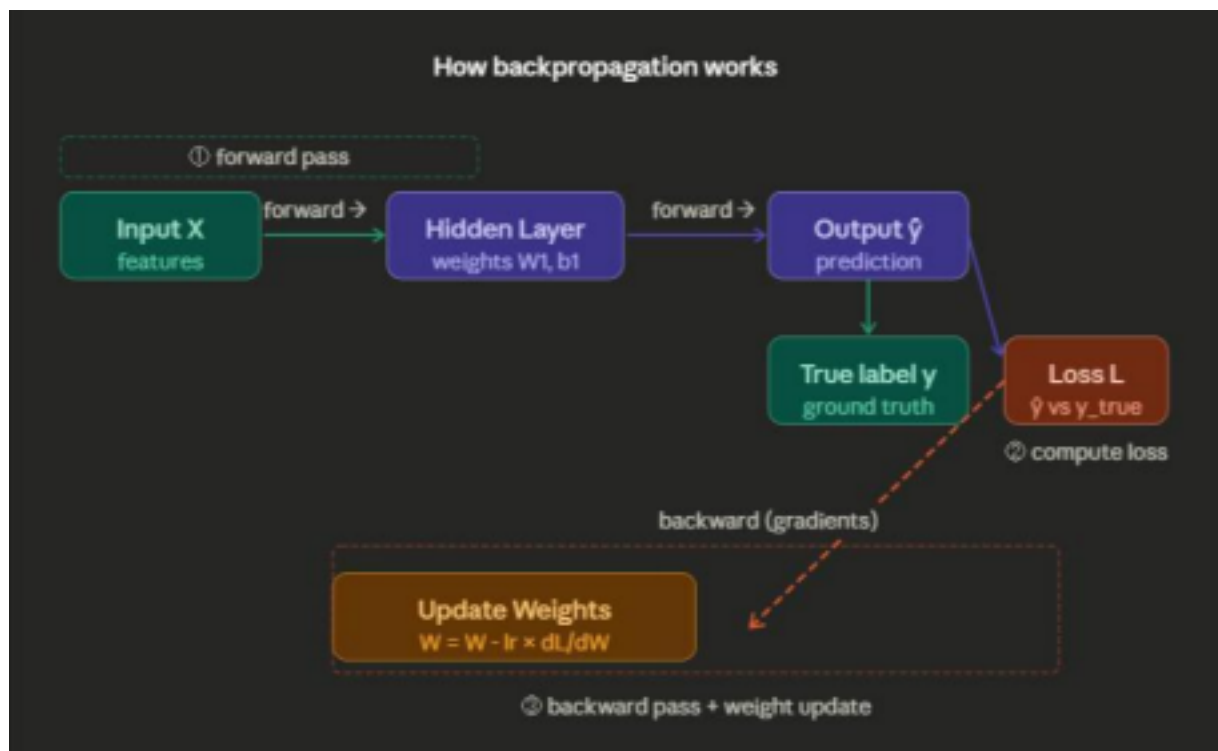
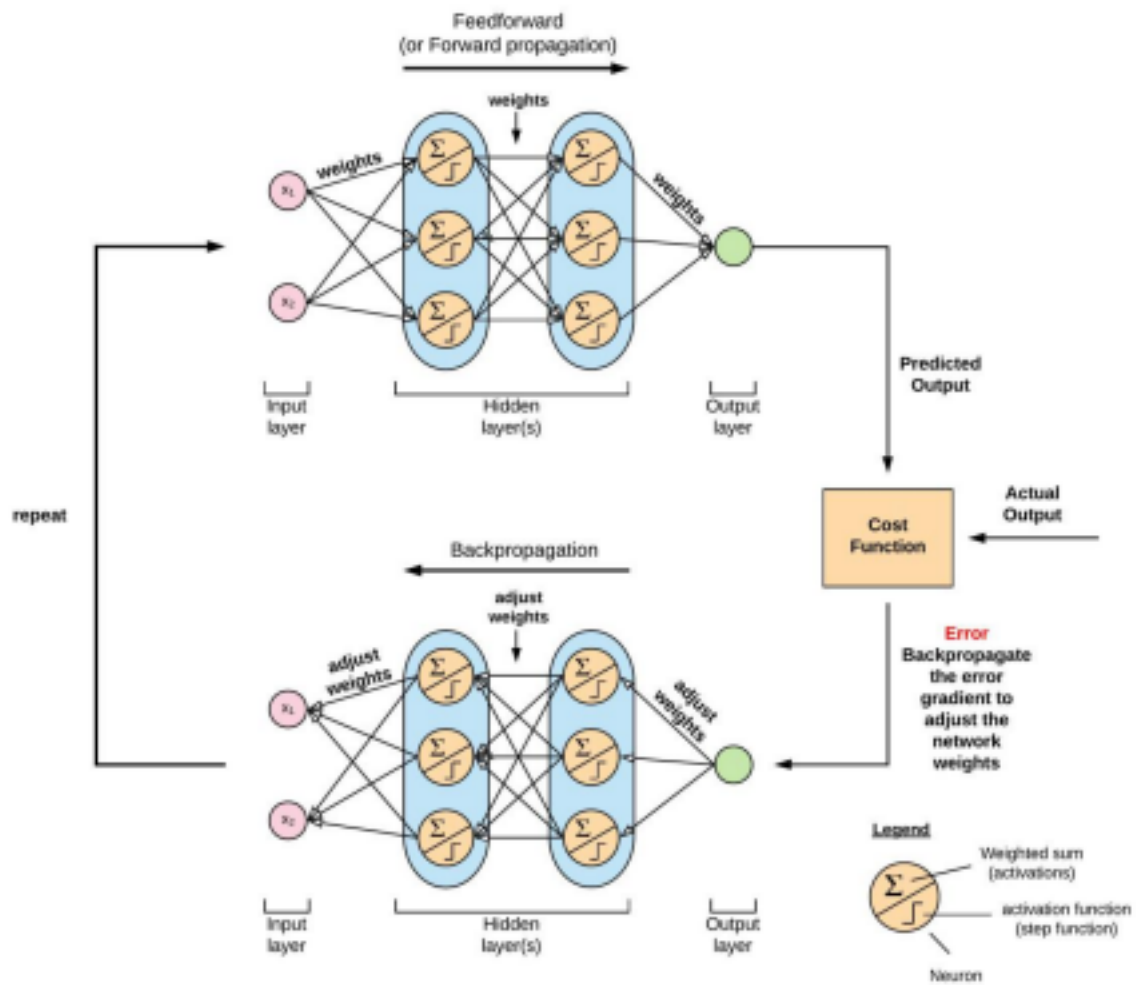
Evaluation Metrics

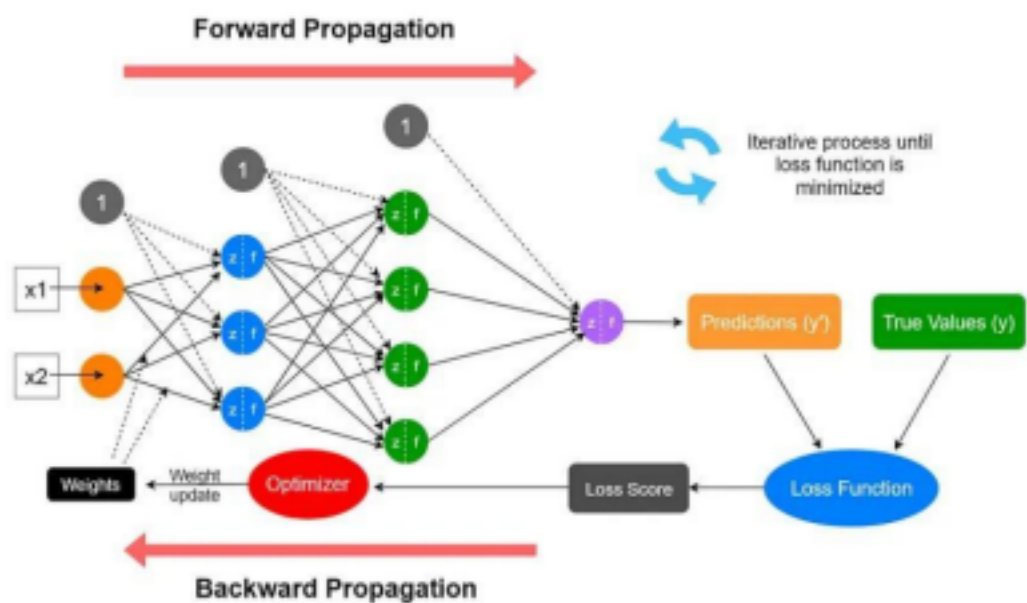
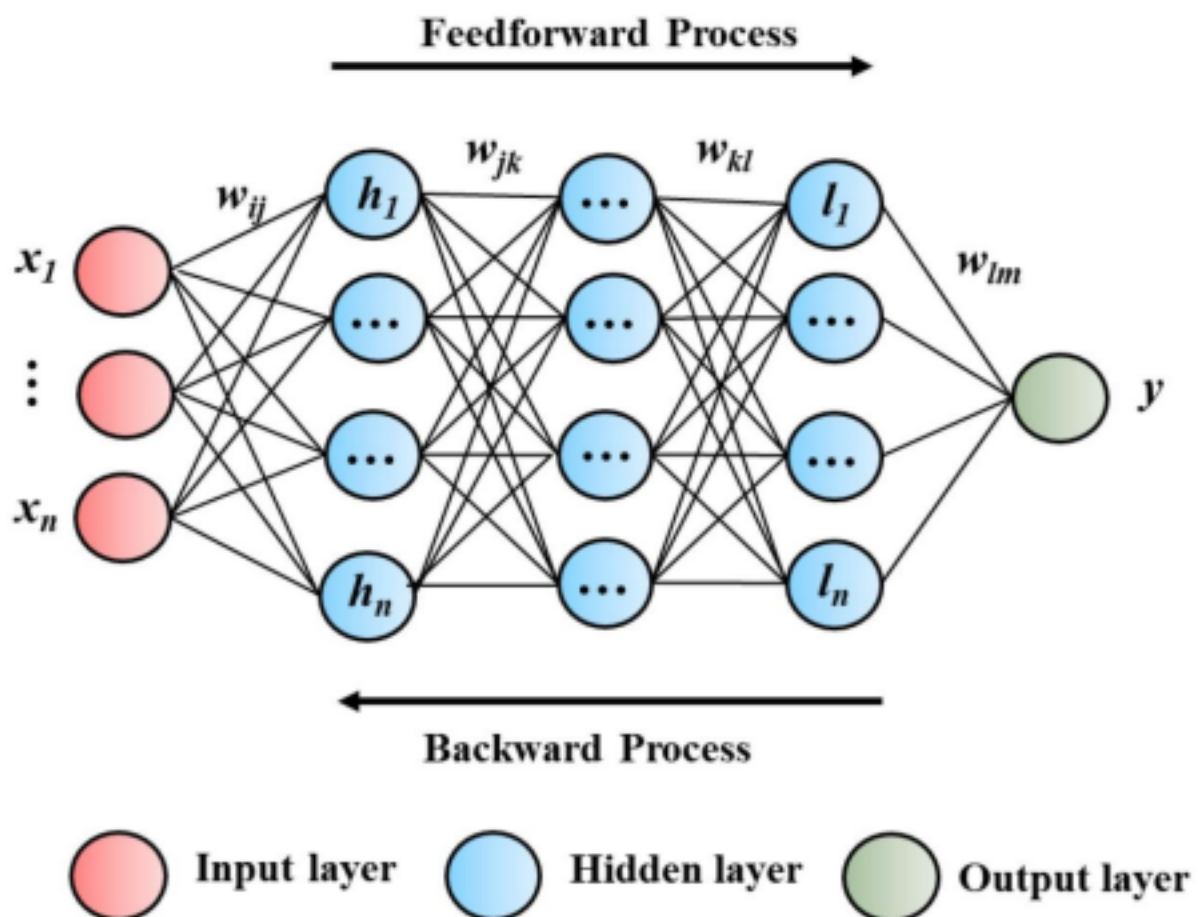
- Accuracy
- Precision
- Recall
- F1-score
- ROC-AUC

Applications

- Early flood warnings
- Earthquake risk prediction
- Disaster risk assessment
- Evacuation planning
- Resource allocation
- Faster emergency response
- Reduced loss of life and property

3. In the context of neural networks, consider a multilayer perceptron used for binary classification. Explain how the backpropagation algorithm updates the weights in the network during training. Illustrate the process using a simple example by describing the forward pass, error calculation, and backward propagation of errors. Discuss how learning rate and activation functions influence the convergence and accuracy of the model.







Multilayer Perceptron (MLP)

Input Layer



Hidden Layer



Output Layer

Backpropagation Process

Step 1: Forward Pass

Input values are multiplied by weights.

Example:

Input = 2

Weight = 0.5

Net = $2 \times 0.5 = 1$

Apply activation function:

Output = Sigmoid(1)

≈ 0.73

Step 2: Error Calculation

Target Output = 1

Actual Output = 0.73

Error = Target - Output

$= 1 - 0.73$

$= 0.27$

Step 3: Backward Propagation

The error is propagated backward.

Weight update formula:

$$\Delta w = e \times a \times w$$

Assume:

- Learning Rate = 0.1
- Error = 0.27
- Input = 2

$$\Delta w = 0.1 \times 0.27 \times 2 = 0.054$$

Updated weight:

$$\diamondsuit \diamondsuit \spadesuit \spadesuit \spadesuit \spadesuit \spadesuit \spadesuit = 0.5 + 0.054 = 0.554$$

Learning Rate

- Controls how much weights are updated.
- High learning rate:
 - Faster learning
 - May overshoot the minimum.
- Low learning rate:
 - Stable learning
 - Slower convergence.

Activation Functions

Common activation functions:

- Sigmoid
- ReLU
- Tanh

Importance

- Introduce non-linearity.
- Improve learning capability.
- Help solve complex classification problems.
- ReLU speeds up convergence, while Sigmoid is commonly used for binary classification.